

Mobile-Optimized Drone Scan Demolecularization

A Step-by-Step Godot Guide for High-Performance Disintegration Effects

This guide covers how to create a mobile-optimized sci-fi drone scan that demolecularizes (disintegrates) an object. This approach uses an alpha discard threshold to delete pixels while adding a glowing emission to the edge, bypassing heavy full-screen post-processing for maximum mobile performance.

Phase 1: The Demolecularization Shader

To achieve the dissolving effect without crashing mobile frame rates, we will use a custom Spatial Shader driven by a noise texture.

1. Create the Spatial Shader

Create a new ShaderMaterial on your 3D mesh, assign a new Shader to it, and paste the following code:shader_type spatial;

```
// Disables culling so we can see the "inside" of the mesh as it dissolves
render_mode cull_disabled, depth_draw_opaque;
```

```
uniform sampler2D albedo_texture : source_color;
uniform sampler2D noise_texture;
uniform vec4 scan_color : source_color = vec4(0.0, 1.0, 0.5, 1.0); // Neon green laser
uniform float scan_progress : hint_range(0.0, 1.0) = 0.0;
uniform float edge_thickness : hint_range(0.0, 0.2) = 0.05;
```

```
void fragment() {
    vec4 albedo = texture(albedo_texture, UV);
    // Read the black-and-white static from our noise texture
    float noise_val = texture(noise_texture, UV).r;

    // Discard (erase) the pixel completely if it falls behind the scan threshold
    if (noise_val < scan_progress) {
        discard;
    }

    // Create a sharp glowing band exactly at the threshold edge
    float edge = step(noise_val, scan_progress + edge_thickness);

    // Apply the base texture, but override it with the neon scan color at the edges
    ALBEDO = mix(albedo.rgb, scan_color.rgb, edge);
    // Force the edge to emit light (glow)
    EMISSION = scan_color.rgb * edge * 2.5;
}
```

2. Configure Node Parameters

Select your object and look at the Shader Parameters in the Godot Inspector. Configure the following slots:

- **Albedo Texture:** Assign your object's standard base color texture.
- **Noise Texture:** Click the slot, select **New NoiseTexture2D**, and assign a **FastNoiseLite**. This creates the randomized "particle" pattern that the mesh will break apart into.
- **Scan Color:** Choose a bright neon color (like cyan, green, or magenta) to match your drone's scanning beam.

Phase 2: Animating the Scan

Shader parameters are static by default. To make the object slowly disappear over time when scanned, you must animate the `scan_progress` uniform.

3. Attach the GDScript Tween

Attach this GDScript to your 3D object to trigger the demolecularization via a

extends `MeshInstance3D`

```
func start_drone_scan():
    # Create a tween to animate the shader's scan_progress from 0.0 to 1.0 over 3 seconds
    var tween = create_tween()
    var material = get_surface_override_material(0)
    tween.tween_method(set_shader_value.bind(material), 0.0, 1.0, 3.0)

    # Optional: Delete the object from the game once fully dissolved
    tween.tween_callback(queue_free)

func set_shader_value(value: float, mat: ShaderMaterial):
    mat.set_shader_parameter("scan_progress", value)
```